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sibling typology: beyond Nerlove and Romney[1]

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In their paper, "Sibling Typology and Cross-Sex Behavior," Nerlove and Romney (1967) introduce a typology of sibling terminologies. They use a number of assumptions based on considerations of cognitive economy and on Greenberg's work on kinship terminological universals to predict which subset of the logically possible types will occur empirically. Their systematic consideration of the full logically possible set of terminological categorizations of siblings and their use of an explicit set of assumptions about the criteria that logically possible terminological categorizations must meet if they are to occur empirically represent impressive contributions to the anthropological study of semantic systems. Equally important is their attempt to derive their assumptions—about the criteria that types must meet in order to exist empirically—from general cognitive and linguistic principles that come from work done on topics other than sibling terms and, where possible, on topics outside of kinship.

However, their specific set of assumptions are not, in fact, sufficient to completely reduce the set of logically possible types to the set of empirically occurrent types, i.e., to eliminate the subset of logically possible types that cannot occur empirically and leave only the subset of logically possible types that do in fact occur empirically. In order to describe my improvement of their analysis I will first briefly characterize their argument and the assumptions they use in it. Next, I will demonstrate which logically possible types are not excluded by their assumptions (even though Nerlove and Romney intend that these types should be excluded). I will complete the major part of the paper by offering improved versions of their assumptions that do in fact exclude these types. My improved versions of their assumptions will be stated in a form that is both more concise than theirs and has a more general range of application than theirs does. In some cases, my versions of their assumptions will be derived more directly and/or more plausibly from the cognitive and linguistic principles or findings that they cite.

These corrections and restatements are important because of the originality and scientific value of Nerlove and Romney's form of analysis. Their deduction of attributes of linguistic categories from experimental findings in cognitive psychology is particularly significant for anthropology because of the many claims that have been made about the

Nerlove and Romney (1967), using assumptions about conjunctivity and markedness, identified a set of necessary conditions which any existent terminological arrangement of true siblings must fulfill. The present paper generalizes and extends their assumptions, explores the psychological bases of these assumptions, and investigates the degree to which these revised assumptions represent a set of conditions such that (given enough cases) any terminological arrangement that meets them should be expected to occur in some natural system.

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independence of cultural and linguistic phenomena from individual psychological phenomena. Also important are their findings concerning the generality of Greenberg's conclusions concerning linguistic universals. If their form of analysis is indeed to provide a model for future work in anthropology, and if their findings are to have general acceptance, it is necessary that the analysis be correctly executed and that their results in fact be entailed by their stated assumptions and adduced data. My revision of their set of assumptions enables these conditions to be met. My improved derivations of my revised assumptions provide the further benefit of making clearer the application of the cognitive and linguistic findings cited by Nerlove and Romney to semantic domains other than sibling terminologies.

I will conclude the paper with a brief discussion of an empirical case in which one might have expected that the Nerlove and Romney assumptions would not hold, but in which they did in fact hold. This discussion will suggest one hypothesis to explain the fact that Nerlove and Romney found only a few cases that apparently did not meet their assumptions.

Nerlove and Romney's analysis

Nerlove and Romney's analysis begins with the fact that true sibling relations are composed of three variables: sex of speaker, sex of relative, and relative age. Each of these variables is independent of the others, and each has two possible values. These three binary variables produce eight (i.e., 2^3) possible kintypes: mome, momy, fome, fomy, mofe, mofy, fofe, fofy (f is a female person; m is a male person; o is a sibling link between two persons; e signifies that the relative is elder than the speaker; y signifies that the relative is younger than the speaker). Nerlove and Romney compute that there exist 4,140 logically possible categorizations of these eight kintypes (cf. Table 1). A categorization is a set of categories (i.e., kinterms) such that each kintype belongs to one and only one (kinterm) category (producing what I call a neat partition of the set of kintypes) and each (kinterm) category contains more than zero but less than nine kintypes. It follows that a categorization may contain anywhere from one to eight categories.

They use an assumption derived from the work of Bruner, Goodnow, and Austin (1956) to eliminate the great majority of these logically possible categorizations (Nerlove and Romney 1967:181). Bruner, Goodnow, and Austin (1956:41-43) found that conjunctively defined categories (characterized by the joint presence of the appropriate values of one or more attributes) are much more easily learned than are disjunctively defined categories (characterized by the individual presence of the appropriate value of at least one of two or more attributes). A category that includes all and only big blue balls is conjunctively defined, while a category that includes all and only balls that are blue and/or big is disjunctively defined. Nerlove and Romney also make some use of Bruner, Goodnow, and Austin's related finding that conjunctively defined categories based on absolute attributes (i.e., where the presence of the appropriate value of the attribute can be determined from the examination of a single object) are somewhat easier to learn than are ones based on relative attributes (i.e., where the presence of the appropriate value of the attribute can only be determined from the comparison of two objects). Color is an absolute attribute; one can tell by a direct examination of a given ball whether or not it is blue. Size is a relative attribute; one needs to compare a given ball with another (or some other equivalent standard) to determine whether the indicated ball is big or small.[5]

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In order to apply Bruner, Goodnow, and Austin's findings to the classification of siblings, Nerlove and Romney "propose to view sibling terminology as the outcome of a 'natural' cognitive experiment, and on this basis assume that disjunctive categories would be extremely rare. This assumes that an 'easier' system has a higher survival value than a 'difficult' system" (1967:181). However, one still cannot classify kinterms as conjunctively or disjunctively defined until one knows what attributes are used to define these kinterms. Presumably as a result of an examination of empirical cases, they determine that there are four semantic features that can function as attributes of sibling terminological categories. Three of these are the variables which they used to define the set of kintypes: sex-of-speaker, sex-of-relative, and relative-age. A fourth, relative-sex (sometimes symbolized as x/ / below), is defined by a comparison of sex-of-speaker with sex-of-relative; this feature has the value "same" (symbolized by / /) if sex-of-speaker is the same as sex-of-relative and "opposite" (symbolized by x) if sex-of-speaker and sex-of-relative are absolute attributes while the features relative-age and relative-sex are relative attributes. As they state, each of these semantic features can be utilized in a given terminology as a *primary* component or distinction (which divides the complete set of eight true sibling kintypes into subsets of four kintypes each), as a *secondary* component or distinction (which divides a subset containing four of the kintypes into two subsets, each of which contains two kintypes), or as a *tertiary* component or distinction (which divides a subset containing two of the kintypes into two subsets, each of which contains a single kintype) (1967:183).

Eliminating all categorizations that contain any disjunctively defined categories eliminates 3,698 of the 4,140 logically possible types of categorizations. These eliminated categorizations include ones such as (mome, fome, mofe, fofy) (momy, fomy, mofy, fofe) and (momy, fofe) (mome, fomy, mofy) (fofy, fome, mofe), in which none of the categories can be conjunctively defined, as well as ones such as (mome, fofe) (momy) (fome, mofe) (fofy, fomy, mofy), in which some of the categories can be conjunctively defined (as, 'elder same sex sibling,' 'younger same sex male sibling,' and 'elder opposite sex sibling,' respectively), but in which at least one of the categories cannot be conjunctively defined (as [fofy, fomy, mofy], the category into which falls 'either a younger same sex female sibling or a younger opposite sex sibling'—'younger sibling' will not work as a definition since momy is in another category).

Nerlove and Romney use five additional assumptions to eliminate the remaining empirically non-occurrent categorizations: (1) Two relative components will not occur as primary; that is, relative age and same-sex/opposite-sex distinctions will not occur together as primary. (2) If sex-of-relative is secondary to relative-age, it will occur in the elder rather than the younger category. (3) If relative-age occurs as a secondary distinction with sex-of-relative primary, it will be on the male side. (4) If relative-age occurs as a secondary distinction, with same-sex/opposite-sex primary, it will be on the same-sex side. (5) Sex-of-speaker may appear as a tertiary distinction (only) (Nerlove and Romney 1967:183).

They seem to have made an additional implicit assumption to the effect that relatively difficult attributes (i.e., semantic features) will be used less completely than will relatively easy ones, i.e., will pertain to fewer categories in a categorization. Such a notion seems necessitated by their explanations of assumptions (1) and (5). Their first numbered assumption is derived from Bruner, Goodnow, and Austin's finding that conjunctively defined categories based on relative attributes are somewhat harder to learn than are ones based on absolute attributes. Nerlove and Romney appear to feel that a categorization

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based on one relative attribute is reasonable but that one based on two is difficult enough to prevent two relative attributes from being primary features (basing their feeling, apparently, on the implicit assumption referred to just above). Assumptions (2) through (5) are derived from Greenberg's work on the theory of marked and unmarked categories (Greenberg, 1966), and especially from the section of that work (Chapter VIII) dealing with universals of kinship terminological systems. In particular, assumptions (2) and (3) are derived directly from Greenberg's general finding that unmarked categories are more likely to be further subdivided than are marked ones (Nerlove and Romney 1967:183). They infer from his findings about kinship universals (his work was not concerned with sibling terms) that elder is unmarked and younger is marked and that male is unmarked and female is marked. Assumptions (2) and (3) then follow as stated. Assumption (4) is modeled on the preceding two, but it is not so certain that relative-sex has marked and unmarked values. Nerlove and Romney's Terminological Types 8, 9, and 12 (see Figure 1) seem to imply that relative sex does not have such marked and unmarked values because these types represent further subdivision of each side of the relative sex feature. Assumption (5) is based on Greenberg's kinship universal which "states that when there is a generational difference between speaker and relative, the sex-of-speaker distinction occurs only with reference to relatives in the lower generation" (Nerlove and Romney 1967:182). They infer from this fact that sex-of-speaker is a difficult feature to learn and support their inference with speculations about an infant's socialization process. This inference, in conjunction with the implicit assumption of theirs that I defined, would preclude sex-of-speaker from being a primary distinction. If the difficulty is assumed to be great enough, it would also preclude it from being a secondary distinction—i.e., would produce their assumption (5). They also suggest, alternatively, that one could extrapolate from Greenberg's universal and say that "sex-of-speaker distinctions are only made with reference to relatives in lower generations and thus will not occur as primary distinctions in sibling terminologies" (1967:182). If "only" excludes the speaker's own generation, then their assumption (5) would be invalid, and so I infer that their extrapolation means that sex-of-ego can only be a feature of younger relatives in speakers own generation and thus that sex-of-ego cannot be a primary distinction. However, this line of reasoning does not explain why sex-of-ego should not be a secondary distinction.

Nerlove and Romney assert that "under the restrictions imposed by the assumptions above, there are just 12 types of combination of primary and secondary distinctions or components. Figure 1 diagrams the 12 types. These 12 types are the major ideal types. Derivative types may be formed with tertiary distinctions" (1967:183).

In this figure, as in later figures and textual illustrations, distinctions are indicated as follows: a vertical line represents sex-of-relative, with males to the left and females to the right; a horizontal line represents relative-age, with elder above the line and younger below; a slanted line from upper-left to lower-right represents relative-sex, with opposite-sex (x) above the line and same-sex (/ /) below; a line from lower-left to upper-right represents sex-of-speaker, with males above the line and females below. Occasionally, where useful, the actual kintypes will be written in.

types omitted in Nerlove and Romney's analysis

There exists, however, a number of logically possible types of terminologies for which Nerlove and Romney's assumptions do not account. These types are not listed among the twelve types that they want to derive and, according to their data, do not occur empirically. In fact, however, these types are not excluded by their assumptions.

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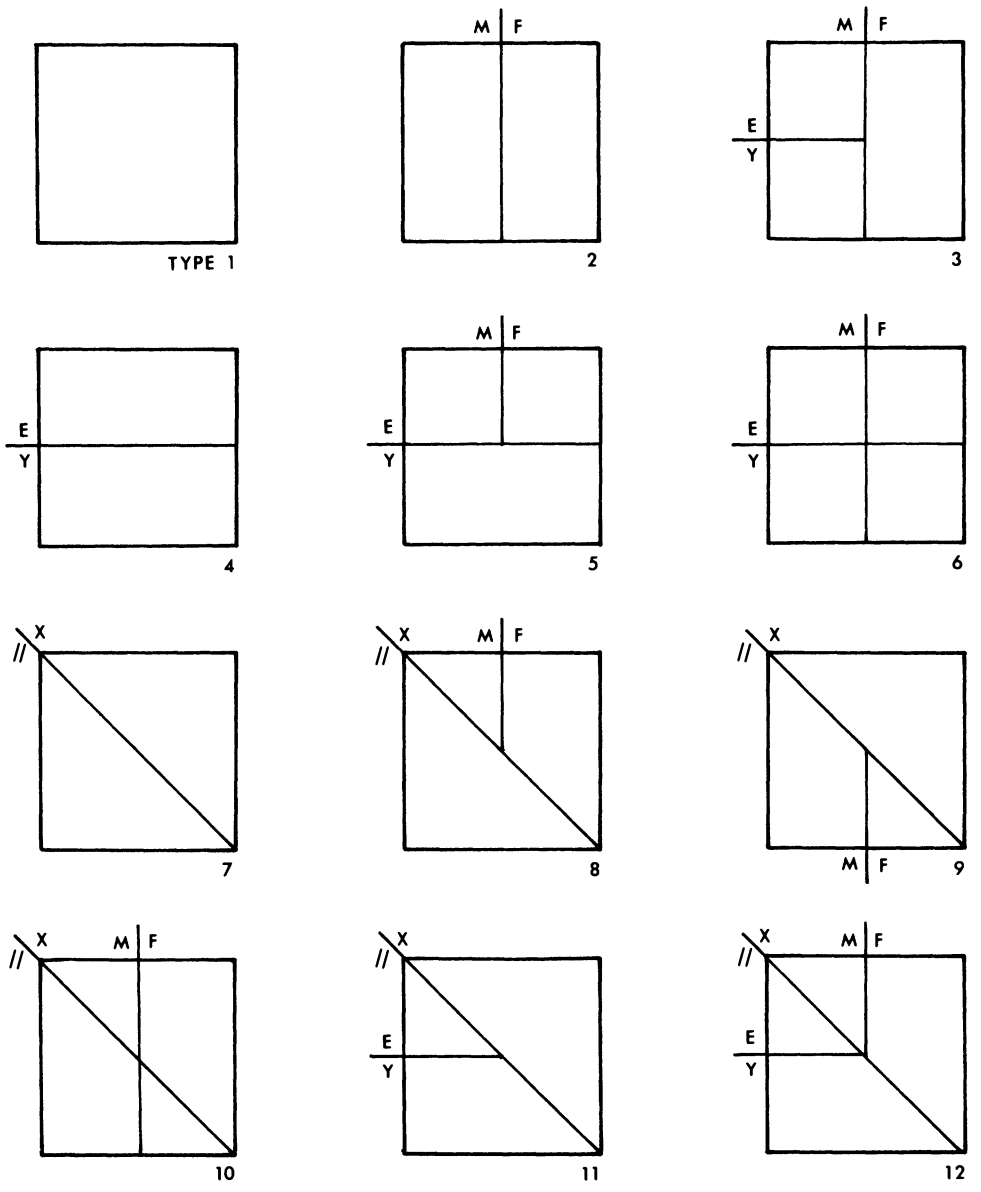


Figure 1. Nerlove and Romney's twelve empirically occurrent types.

Type I. They ignore a great number of types (206 by my count) which use sex-of-elder-sibling or sex-of-younger-sibling as distinctions (i.e., as distinctive features). These two features account for the other two categorizations (of the six Nerlove and Romney say exist) that would divide the set of eight sibling kintypes into two conjunctive kinterms of four kintypes each (aside from the four they explicitly treat, based on the four relevant features named above).

Type II. They do not consider, but also do not exclude, the four types formed by using x/ / as a secondary distinction that are shown in Figure 2.

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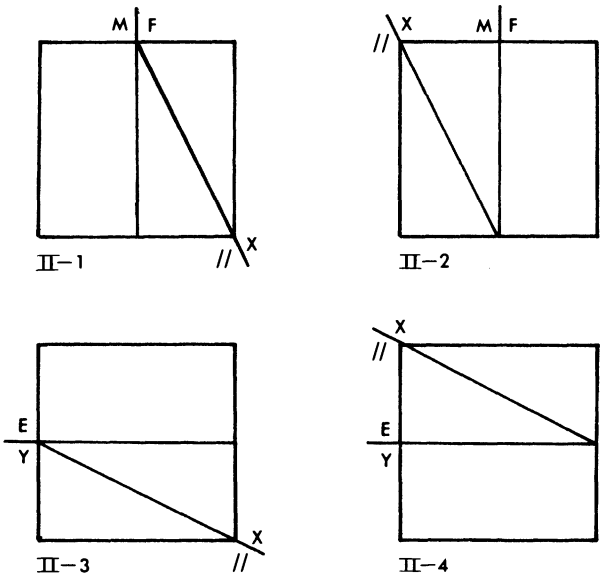


Figure 2. Four types using cross/parallel as a secondary feature.

Type III. They claim to consider all "types of combinations of primary and secondary distinctions or components" (Nerlove and Romney 1967:183). In this case there are a number of types which their assumptions do not preclude. One group of such types is constructed by using two secondary distinctions on the same side of a primary distinction (see Figure 3). Another group is constructed by using two primary distinctions together with one secondary distinction (see Figure 4). A final type is constructed by using three primary distinctions (see Figure 5). There are in all sixteen types formed in these three ways which are not exluded by their assumptions. I will treat these types later, but I would like meanwhile to suggest that Nerlove and Romney really misstated their intention. I think that they really meant to define "major ideal types" as those which contain at least two kintypes per category—i.e., in terms of Table 1, 8-0, 4-4, 2-2-4, and 2-2-2-2 divisions or categorizations. Since Figure 3 contains types with a 1-1-1-1-4 division, Figure 4 types with a 1-1-1-1-2-2 division, and Figure 5 a type with a 1-1-1-1-1-1-1-1 division, the types shown in these figures would fall outside their limited universe of "major ideal types." I will later discuss the effect on these types of my versions of the Nerlove and Romney assumptions.

Type IV. They explicitly do not use their assumptions to predict which types might be expected to occur with the use of tertiary distinctions. But, in fact, my restated

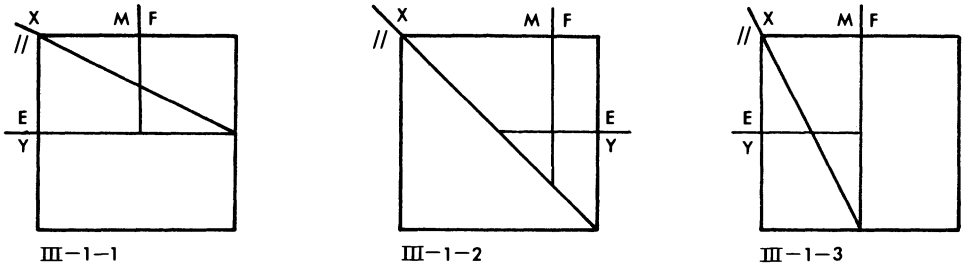


Figure 3. Types with two secondary distinctions on one side of a primary distinction.

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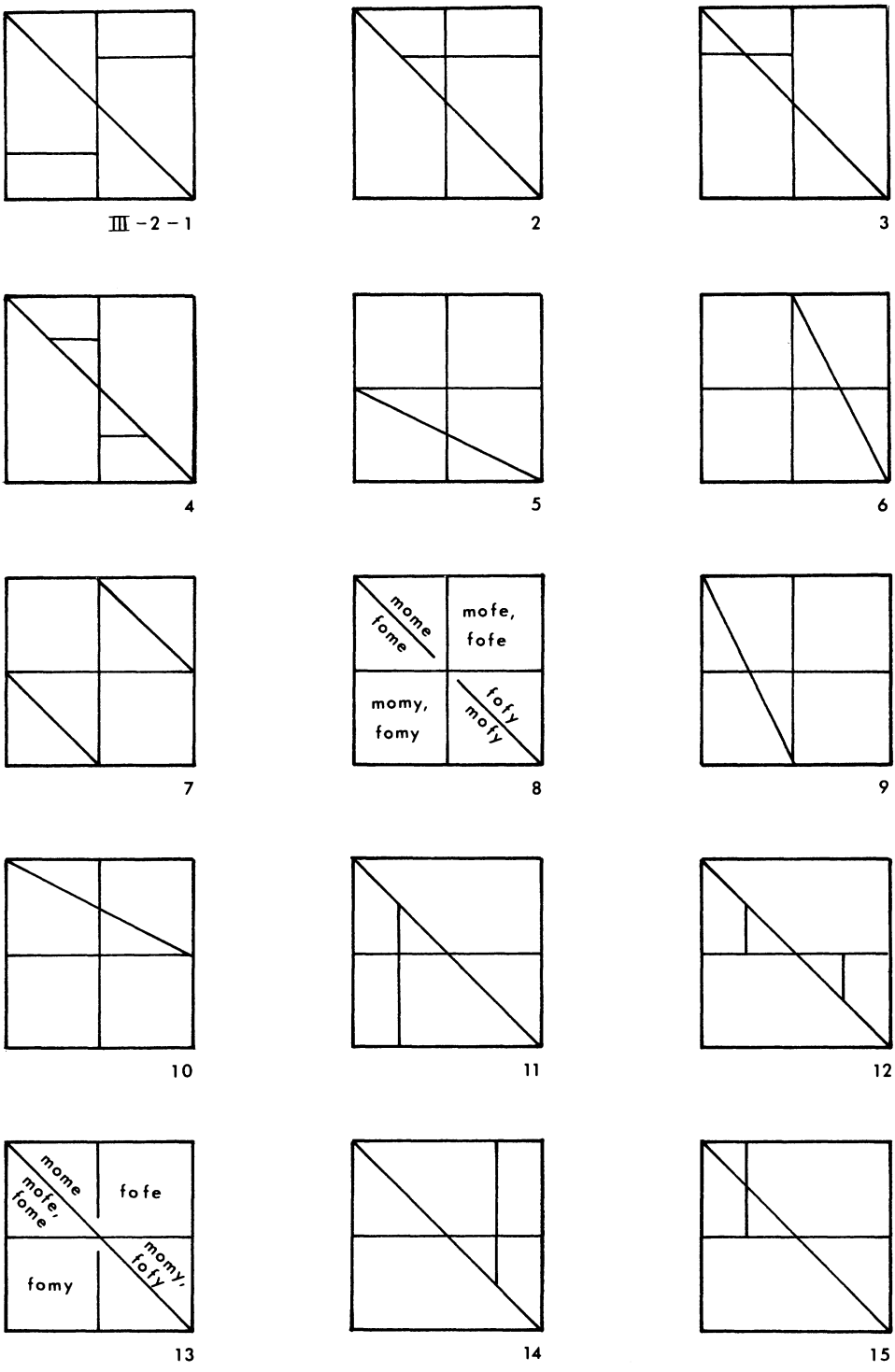
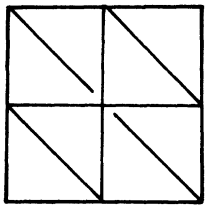


Figure 4. Types with two primary distinctions and one secondary distinction. All are type III-2.

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III—3—1

Figure 5. Type using three primary distinctions.

assumptions enable a great many such logical possibilities to be eliminated. I will indicate these eliminations, even though the data Nerlove and Romney provide do not allow us to evaluate the accuracy of these predictions. Those types using tertiary distinctions which I predict to occur are shown later (in Figure 6) since they make no predictions about such types.

I will now examine their particular assumptions in order to see how these may profitably be restated and/or condensed. Following that restatement, I will suggest two further assumptions that are required by the Type I and Type II situations above. Assumption 1, that two relational components will not occur as primary components, is a little unsatisfying: no reason is suggested why cognitive economy (their basis for the assumption) would preclude two relational features from being primary components, but would not preclude a situation such as their Type II, in which a primary relative component occurs with a secondary relative component. One might be tempted to broaden this assumption to include the case of two secondaries (in order to eliminate my Type III-1-3), but it seems preferable not to do so because of the ambiguity about just what generates the assumption. Other, more satisfying, reasons for eliminating III-13 will be given below.

So, let us keep their assumption 1 as it stands and combine their assumptions 2 (if sex-of-relative is secondary to relative age, it will occur in the elder rather than the younger category) and 3 (if relative-age occurs as a secondary distinction with sex-of-relative primary, it will be on the male side) into a new assumption 2: relatively incompletely extended features will only occur on the unmarked side of relatively completely extended ones. Thus, secondary features should only occur on the unmarked side of primary features, and tertiary features should occur only on the unmarked side of primary or secondary features. Male and elder are assumed to be unmarked relative to female and younger, respectively. There appear to be no marking differences between same-sex and opposite-sex. This revised assumption eliminates my Types II-1 and II-3, above, and my Types III-2-1, 4, 5, 6, 7, 8, 12, and 13.

Nerlove and Romney offer no psychological basis for their assumption 4, namely, that if relative-age occurs as a secondary distinction with sex-of-relative primary, it will be on the same-sex side. Notions of cognitive economy would seem to imply that terminological distinctions are not made without a reason. I would like to suggest a hypothesis (though at present without proof) about non-self-reciprocal kinterms which, in conjunction with cognitive economy, would account for their assumption 4. If non-self-reciprocal sibling terms imply differences in general status, then for opposite-sex siblings sex-of-relative defines the status, with males being above females; same-sex siblings, on the other hand, are symmetric with regard to sex, and thus differential status can only (in the context of our present terminological game) be based on relative-age. Sex-of-relative distinctions on the same-sex side would have to derive from specific relations such as succession to some property or office. Thus, we have an hypothesized (and testable) functional basis for their assumption 4 (if relative-age occurs as a secondary distinction, with same-sex/opposite-sex

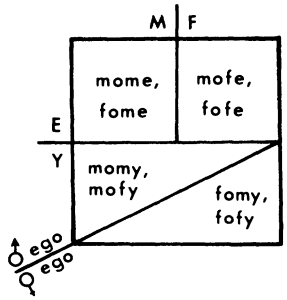
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primary, it will be on the same-sex side): sex-of-relative as a secondary distinction on the opposite-sex side implies the presence of general status-related behavior, while sex-of-relative as a secondary distinction on the same-sex side implies the presence of succession to some particular item.

Their assumption 5 (sex-of-speaker may appear as a tertiary distinction) is based on Greenberg's (1966:78-79) discussion of sex-of-ego as a kinterm feature. Greenberg is explicitly talking about generations other than G^0; he says that sex-of-speaker occurs as a feature of descending generation terms that are true-reciprocals of ascending terms based on sex-of-relative. (Psychologically, the distinguishing feature of the descending term, $\bar{x}$, seems to be "reciprocal-of-x" rather than "sex-of-speaker.") The only exact analog in G^0 would be the case of relative-age as a primary feature, where older siblings were distinguished by sex-of-relative (as a secondary feature) and younger siblings, reciprocally, by sex-of-speaker (as a secondary feature), a case which does not occur in Nerlove and Romney's data.

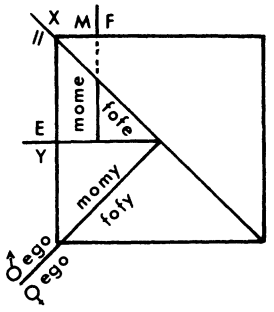
Greenberg's findings seem to depend on a situation in which the senior relative has his social position clearly established before the junior enters on the scene; his position has to be different from the one the junior initially assumes upon appearing, but it also has to be one in which the junior has a long-term interest. The kind of relationship being discussed is focused particularly on the senior relative who has the position, rather than being focused on the relationship between the two relatives, as are most other terms. For example, with the Fanti, a+fom (*wofa*) is the person from whom I inherit; when I am born, he already has property and position in the lineage. On the other hand, mof-a (*awofasi*) has nothing and is simply the person who inherits from me. The Fanti seem more concerned with the role of the *wofa* than with that of the *awofasi*. Older siblings are not usually as clearly established at the birth of younger siblings as are relatives of ascendent generations; therefore, it seems unlikely to me that Greenberg's findings could be used as the basis for inferring that sex-of-speaker is ever used as a distinctive feature of sibling terminologies.

Having seen a general reason for not expecting sex-of-speaker ever to be a distinctive feature of sibling terminologies, let us next examine what use Nerlove and Romney make of this feature in their sibling terminology typology. Their reading of the implications of Greenberg's findings for sibling terminologies excludes all uses of sex-of-speaker except those in which younger sibling terms generated by sex-of-speaker are true reciprocals of elder sibling terms generated by sex-of-relative. Any use of sex-of-speaker as a primary distinction is obviously ruled out. The only possible type in which sex-of-speaker could be a secondary distinction is the one just discussed in which relative age is the primary distinction, sex-of-relative is the secondary distinction on the elder side, and sex-of-speaker (reciprocally) is the secondary distinction on the younger side.



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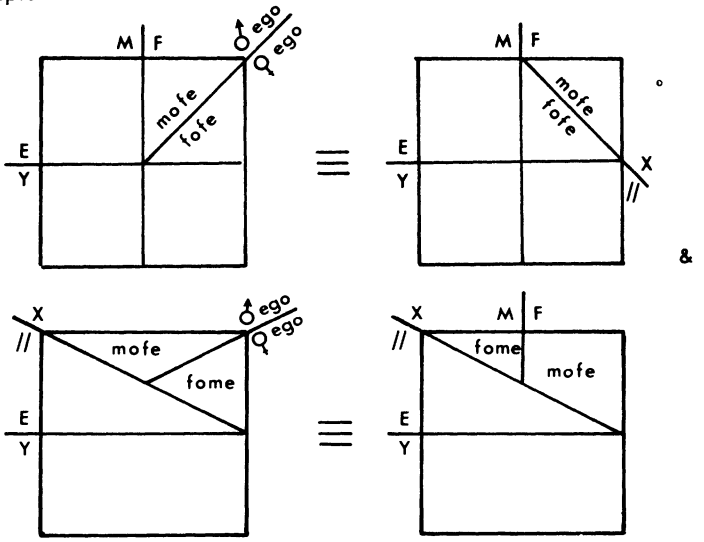
This type does not occur in their empirical data concerning occurrent types; therefore, they want to eliminate it. Similarly, there is only one possible type in which sex-of-speaker could appear as a tertiary distinction. This type is based on Nerlove and Romney's Type 11; in it, elder siblings are distinguished by sex-of-relative and younger same-sex siblings by sex-of-ego.



Nerlove and Romney found only two actually occurring terminologies which used any kind of tertiary distinctions to subdivide Type 11. They do not tell us what tertiary distinctions were used in either case; the type just described does not represent the only possibility. Thus we can see that, using their reading of Greenberg, Nerlove and Romney keep sex-of-speaker as a possible tertiary distinction only for the sake of one type that may not even occur. On this basis, too, I think one might conclude, as I did above on other grounds, that sex-of-speaker should not be expected to be a feature of empirically occurrent sibling terminologies.

However, their assumption 5 does not limit itself to the situation implied by their reading of Greenberg; it simply states that sex-of-speaker may [only] appear as a tertiary distinction [anywhere consistent with their other assumptions]. (The bracketed words are implicit in their statement.) However, an examination of the actual types makes clear that any categorization of the eight kintypes which can be made using sex-of-speaker as a tertiary distinction can also be made using either sex-of-relative or relative-sex as the tertiary feature in its place.

For example:



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Sex-of-speaker, if limited to appearing only as a tertiary distinction, contributes nothing and so can be dispensed with.

Thus, their assumption 5 should be replaced with an assumption that sex-of-speaker simply will not be a feature of sibling terms. I think that a consideration of cognitive economy indicates some further reasons for ruling out sex-of-speaker as a feature in anything but the true-reciprocal or self-reciprocal situation.

Bruner et al. suggest that we handle cognitive problems best where we can have simple *if-and-only-if* definitions of categories in terms of single sets of features. Conjunctive categories seem to be preferred over relative categories because the former consist wholly of absolute features that require only a single observation for evaluation, while the latter necessitate more work, requiring a comparison of two observations for categorization. This is the basis for Nerlove and Romney's assumption 1. The actual observations that a child would have to make to form the sex-of-speaker concept (other than in the reciprocal situation discussed above) would seem to be mildly complex, involving the conception of oneself as a classifiable concept in the world. This "decentering" (or the realization that the world does not exist only from the one perspective of one's own self) is something that Piaget and others think a child has trouble doing at the age at which he would be learning sibling terminological categories (cf. Flavell 1963:230-231). This argument about cognitive complexity completes the reasoning behind my version of assumption 5. How a child perceives his kinsmen and learns his kinship categorizations is a relevant consideration for this argument: one presumes that it is through the process of transmission from generation to generation that the individual's tendency to infer regular conjunctive attributes from the set of instances to which he is exposed comes to affect an interpersonal structure such as language (also here cf. Brown 1965:247)—Nerlove and Romney's " 'natural' cognitive experiment."

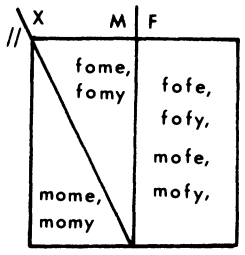
Having finished revising Nerlove and Romney's assumptions, we can now proceed to my two additional assumptions. We saw above that there are a number of types (my Type I situation) based on the features, sex-of-elder-sibling and sex-of-younger-sibling, which are not among Nerlove and Romney's twelve empirically occurrent types, but which are not eliminated by their explicit assumptions. These two features would be even more complex than sex-of-speaker since they first require the comparison of two observations (as does a relative feature), but then also require a subsequent observation, not of an absolute thing, but of the result of the first comparison. It *does* get hard to keep straight; Bruner et al.'s work on concept formation strategies (1956:Chapter 4) shows that people prefer strategies that are relatively easy to keep straight. On this basis we can add a new assumption: (6) sex-of-elder-sibling and sex-of-younger-sibling will not occur as features of sibling terms.

Let us now turn to the situation represented by the use of x/ / (relative-sex) as a secondary (or tertiary?) distinction. Nerlove and Romney propose no such type in their list of twelve types and indicate that no such types occur empirically. However, they offer no assumption that would rule out such types. I suggest, as a new assumption (7), that relative-sex has logical properties involving self-reciprocality that prevent it from being used as a secondary distinction or, in all but one special case, as a tertiary distinction. The special case is the separation of mome from fome in Nerlove and Romney's Types 3, 5, and 6.

The symbol x/ / represents a judgment (relative to some cognitive or behavioral purpose) that one's own-sex siblings are significantly more similar to one than are one's opposite sex siblings. One such situation, cross-sex sibling avoidance, is discussed in the article: the relatively weak correlation that they found between this avoidance and the

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existence of the x/ / feature suggest that the same/different distinction must be made for other purposes, also. However, any such same/different basis that had any behavioral consequences would prevent x/ / from appearing as a secondary feature: let us look at my Type II-2 as an example. If sex-of-alter were the primary, a woman would, in the avoidance case, not avoid her sister (fof), nor would a man avoid his sister (mof), since all



sisters are in the same category. A man would, similarly, not avoid his brother (mom), but, since a woman's brother (fom) was in a different category, she would avoid him, producing the anomalous situation of a woman avoiding her brother while he did not avoid her. Alternatively, taking the argument from behavior to terminology would imply that, if brother-sister avoidance led to fom being distinct from mom, then it would also lead to the distinction of mof from fof. A similar argument has Ego avoiding alter for bobe[3] but not for its reciprocal, boby, in the other type (II-4) in which one might expect x/ / to be a secondary feature. Presumably, this logical problem would also occur with same/different judgments based on relations other than avoidance. Therefore, a true x/ / distinction would seem only to be able to be a primary distinction, eliminating Types II-2, II-4, III-1-1, III-1-3, III-2-9, and III-2-10. (Note that Types II-1 and II-3 have already been eliminated by our revised assumption 2.)

I do, however, have one speculative hypothesis about a situation in which one might want to speak of something like x/ / as a tertiary distinction. By elimination, in the tertiary case, sex-of-relative and relative-age would have to be primary and/or secondary features. Subsequent subcategorization, by assumption 2, can only occur on the unmarked side of each of these. Hence x/ / as a tertiary can only be used to separate mome from fome (cf. Nerlove and Romney's Types 3, 5, and 6). If any special status or position goes with holding the position of senior sibling (where sex-of-relative is coded also), it will devolve from senior male to junior male. A male might use a special term to the person, mome, from or through whom he expected to receive the position. For a female, her older brother is in a different category: one implying high status, but one not implying any potential inheritance of the position. She does not, however, have a special term for her sister because there is no special female position. Thus, mome would be differentiated from fome, while mofe would not be separated from fofe nor momy from fomy. To evaluate my hypothesis about this one possible occurence of x/ / as a tertiary one would need to know just what societies accounted for the tertiaries found *re* Nerlove and Romney's Types 3, 5, and 6 and look at the indicated information for them. Approaching the problem from the search for such axioms does allow one to frame some very specific hypotheses, which would be very interesting if it proved to be correct.

apparent errors: disjunction or overlapping terms

I would like to conclude this paper with a relevant observation about the Fanti of Ghana, whom I studied in 1967-1968. The Fanti categorization of siblings does not

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constitute a neat partition of the sibling set: one kintype falls into two kinterms. For this reason, the Fanti categorization falls outside of the universe of 4,140 types of categorizations that Nerlove and Romney begin with. However, it violates none of the assumptions that they use to eliminate empirically non-occurrent types from that universe. An examination of this Fanti categorization offers some insight into the strength of Nerlove and Romney's basic assumptions.

If one assumes, with me, that distinctions get coded in a terminology for specific reasons (even if we do not yet have a good theory about these reasons), then one can ask what happens when the reasons seemingly operate so as to imply a disjunctive terminology. In particular, there are two things that can happen: (1) the partition of the sibling set that Nerlove and Romney assume (in which each sibling type belongs to one and only one sibling term) can break down, or (2) the conjunctivity assumption can break down. The Fanti apparently have such pressures toward disjunctivity: for most purposes, siblings are sufficiently similar not be terminologically distinguished, but mof must be distinguished because she is a man's terminological link to his matrilineal heirs (his "mof-a" is his "mof's child") and is required for the relative product calculus by which Fanti calculate kinship (Kronenfeld n.d.; cf. Wallace 1970:841-845). The disjunctive, but neatly partitioned, solution, which is the one I would have predicted prior to reading Nerlove and Romney's article and prior to seeing the Fanti, would be for them to have terms: (1) mof and (2) mom, fom, fof. The conjunctive, non-neat-partition, solution that the Fanti in fact have chosen also requires two terms: (1) mof and (2) aoa (i.e., mom, fom, fof, and mof). The Fanti clearly are preserving conjunctivity—even at the price of having overlapping terms. The Fanti case, in turn, makes me curious about the five disjunctive cases reported by Nerlove and Romney; could, perhaps, the ethnographers involved have missed similar overlapping kinterms; will assumption (1) always break down before (2) or are there genuine disjunctive categories in which (2) breaks down first? Faris (1969) has suggested that an apparently disjunctive (sixth) case is really a polysemous case involving sex-of-ego as a feature. I suggest, on the basis of the material he presents, that his Nuba case is essentially the same as my Fanti one: a male's sister is that male's terminological connection to his matrilineage descendents and is specially labelled as such, but in other contexts is simply a sibling and labeled by the general sibling term. Faris's statement that "mixed siblings are *laŋ*," the general sibling term (1969:483), provides some support for this view. But whether I am right or wrong about Nuba, both Faris's observations about that case and mine about Fanti indicate the need for additional ethnographic study, in view of Nerlove and Romney's findings, of the five apparently disjunctive cases that they found in the literature. The alternatives that Faris and I suggest have both not normally been considered by fieldworkers studying kinship systems until recently.

summary

I would now like to recapitulate briefly the status of the corrected sibling typology. Table 1 shows the number of kintypes logically possible. It reproduces Nerlove and Romney's Table 1 with some additional data about the number of conjunctive categories of various kinds. Table 2 indicates the effects of the basic assumptions (conjunctivity and my assumption 6 concerning sex-of-elder and sex-of-younger as features) on the data of Table 1. Table 2 also indicates, for the same data, the distribution of conjunctive vs. relational categorizations and the distributions of categorizations using primary, secondary, and tertiary distinctions. Table 3a compares the effects of Nerlove and

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Table 1. Number of kintypes per term logically possible.*

Number of Terms	Distribution of Kintypes per Term				
1	8 (1)[1] 0:1 1:0:0:0				
2	1-7 (8)[0]	2-6 (28)[0]	3-5 (56)[0]	4-4 (35)[6] 2:4 0:4:0:0	
3	1-1-6 (28)[0]	1-2-5 (168)[0]	1-3-4 (280)[16] 16:0	2-2-4 (210)[27] 9:18 0:0:18:0	2-3-3 (280)
4	1-1-1-5 (56)[0]	1-1-2-4 (420)[50] 18:32 0:0:0:32	1-1-3-3 (280)[8] 8:0	1-2-2-3 (840)[32] 32:0	2-2-2-2 (105)[22] 6:16 0:4:12:0
5	1-1-1-1-4 (70)[12] 4:8 0:0:8:0	1-1-1-2-3 (560)[64] 64:0	1-1-2-2-2 (420)[88] 20:68 0:0:0:68		
6	1-1-1-1-1-3 (56)[8] 8:0	1-1-1-1-2-2 (210)[89] 17:72 0:0:16:56†			
7	1-1-1-1-1-1-2 (28)[18] 2:16 0:0:0:16				
8	1-1-1-1-1-1-1-1 (1)[1] 0:1 0:1:0:0				

Key:
Numbers separated by dashes indicate the number of kintypes per term.
Numbers in parentheses indicate the number of unique combinations.
Numbers in brackets indicate the number of conjuntively defined categoriations.
First ratio compares the number of conjunctively defined categorizations using sex-of-elder or sex-of-younger as features with the number using only the other four features.
Second ratio divides the second number from the first ratio according to how many of the categorizations use no distinctions, only primary distinctions, only primary and secondary distinctions, and all three kinds, including tertiaries.

*Taken from Nerlove and Romney's Table 1 (1967:180). I have added the bracketed information and the ratios.

†The ratio of secondaries to tertiaries in this case is not automatically certain; it depends, in part, on the order in which one considers alternative arrangements of the eight kintypes in terms of features and, in part, on how some ambiguous cases are assigned.

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Table 2. Summary of information from Table 1.

	Number of terms								Totals	
	1	2	3	4	5	6	7	8		
Total number of unique combinations	1	127	966	1701	1050	266	28	1	4140	
Disjunctively defined	0	121	923	1589	886	169	10	0	3698	
Conjunctively defined	1	6	43	112	164	97	18	1	442	4140
Sex-of-elder and sex-of-younger used as features	0	2	25	64	88	25	2	0	206	
Sex-of-elder and sex-of-younger not used as features (my assumption 6)	1	4	18	48	76	72	16	1	236	442
Only absolute attributes used	1	2	4	1	0	0	0	0	8	
⩾1 relative attribute used	0	2	14	47	76	72	16	1	228	236
No distinctions made	1	0	0	0	0	0	0	0	1	
Primary distinctions only used	0	4	0	4	0	0	0	1	9	
Primary and secondary distinctions used only	0	0	18	12	8	16	0	0	54	
Primary, secondary, and tertiary distinctions used	0	0	0	32	68	56	16	0	172	236
⩾2 kintypes per term	1	4	18	16	0	0	0	0	39	
Other	0	0	0	32	76	72	16	1	197	236

Table 3. Comparison of effects of Nerlove and Romney assumptions 1-5 with effects of Kronenfeld assumptions 1, 2, 4, 5, and 7 on the universe of conjunctively defined categorizations which do not use sex-of-elder and sex-of-younger as features (cf. my assumption 6).

						Empirically Existent Types	Others Not Excluded	Totals
Nerlove and Romney								
Assumption #, by which categorizations are excluded from empirical universe	5	1	2	3	4			
Categorizations with all categories containing two or more kintypes	14	1	2	2	2	12	6	39
Categorizations with at least one category that contains only one kintype	8	0	2	2	2	0	18	32
Totals	22	1	4	4	4	12	24	71
Kronenfeld								
Assumption #, by which categorizations are excluded from empirical universe	5	1	2	4	7			
Categorizations with all categories containing two or more kintypes	14	1	8	2	4	12	0	39
Categorizations with at least one category that contains only one kintype	8	0	12	2	2	0	8	32
Totals	22	1	20	4	4	12	8	71

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Romney's assumptions 1-5 with my assumptions 1, 2, 4, 5, and 7 on the universe of thirty-nine possible types (in which all categories possess at least two kintypes) that remain after eliminating disjunctive types and types using sex-of-elder and sex-of-younger as features. Table 3b makes the same comparison for the remainder of the universe of types formed exclusively by combinations of primary and secondary distinctions or components. Figure 6 presents the 1-1-1-1-4, 1-1-1-1-2-2, and 1-1-1-1-1-1-1-1 types which my revised assumptions do not eliminate, and thus which are predicted to occur empirically. Figure 7 similarly presents the only types of terminologies using tertiary distinctions that are not excluded by my revised assumptions.

The assumptions of Nerlove and Romney with the refinements and additional assumptions of mine represent a set of necessary conditions for the empirical existence of a type of sibling terminology. The set of assumptions presented in this paper comes close to being a maximally possible such set since the assumptions apparently eliminate all empirically non-existent types which contain no fewer than two kintypes per category. There are possibly some cases of non-elimination or of over-elimination represented by types in which one or more of the categories contains only one kintype; Nerlove and Romney did not provide us with data about empirical cases for such types. Most of the assumptions are derived in a straightforward manner from simple notions of cognitive economy, especially notions developed by Bruner, Goodnow, and Austin. Other assumptions derive from Greenberg's work on marked and unmarked categories and from

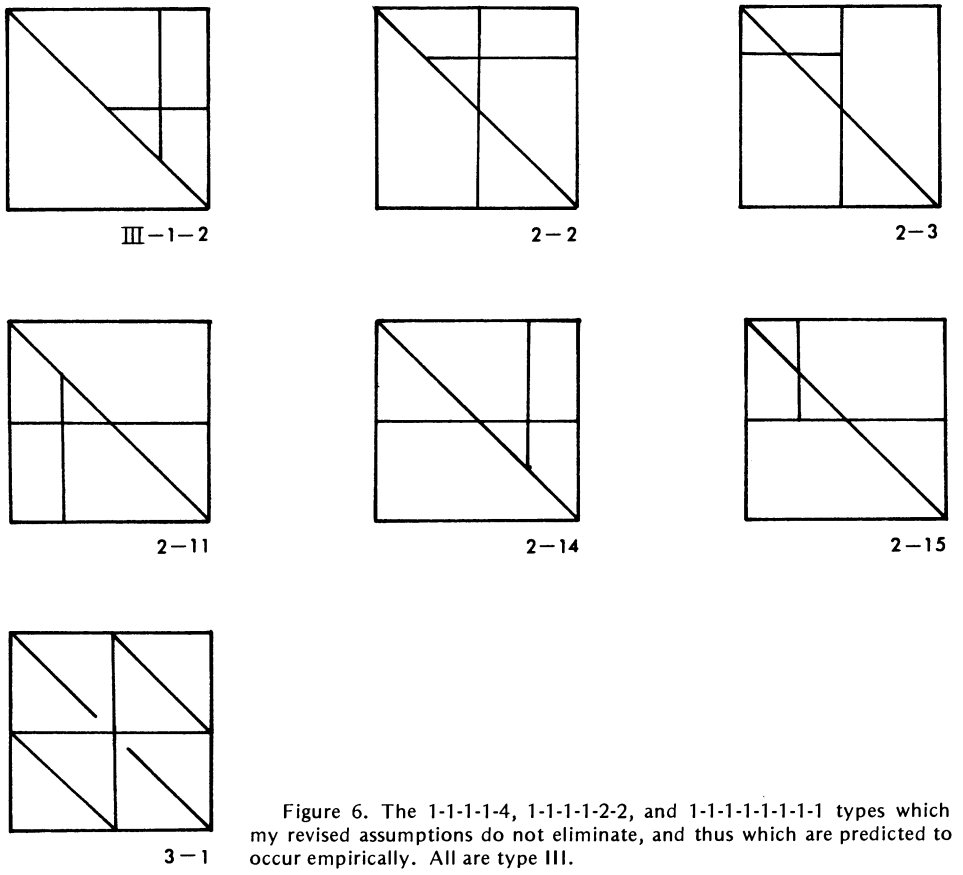


Figure 6. The 1-1-1-1-4, 1-1-1-1-2-2, and 1-1-1-1-1-1-1-1 types which my revised assumptions do not eliminate, and thus which are predicted to occur empirically. All are type III.

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Number of Kintypes per Category	Major Ideal Type	Tertiary Type Derived from Major Ideal Type	Reasons and Comments
	1,2,4,7	none possible	None of these contain any categories that have only 2 kintypes that could be divided by a tertiary distinction.
	IV-1		
1-1-2-4	6,10,2	none possible	None of them have any categories that contain 4 kintypes as required by the division.
	3	IV-1-1 	See special case discussion *re*: assumption 7.
	5	IV-1-2 	See special case discussion *re*: assumption 7.
	8	IV-1-3 	If assumption 4 is not read as also barring the tertiary relative age distinction from the x side of relative sex.
	9	IV-1-4 	
	11	IV-1-5 	
	IV-2		
1-1-2-2-2			
	3,5,8,9,11	none	Each has a category containing 4 kintypes which cannot be divided by a teritary distinction.
	6	IV-2-1 	See special case distinction *re*: assumption 7.
	10	IV-2-2 	
		IV-2-3 	
	12	IV-2-4 	
		IV-2-5 	
	IV-3		
1-1-1-1-2-2			
	3,5,8,9,11	none	cf. IV-2
	6,10	none	Each uses two primary features leaving only one feature to subdivide; could only be done with use of secondary distinction.
	12	IV-3-12 	
	IV-4		
1-1-1-1-1-1-2			
	3,5,8,9,11	none	cf. IV-2
	6,10	none	would require at least added secondary distinctions.

Figure 7. The only types of terminologies using tertiary distinctions which are not excluded by my revised assumptions.

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some simple logical concerns. Only Nerlove and Romney's assumption 1 remains inadequately explained, and it only accounts for the elimination of a single type. The work from which this paper's assumptions are derived applies to semantic systems in general, not just to sibling terms. The assumptions of this paper can be translated into other domains, although one must independently determine the attributes or distinctive features of any given domain, and one must determine what marking relations, if any, hold in that domain. One would hope that empirical data in other domains would provide the same support for these or related assumptions that the sibling terminology data provides. Nerlove and Romney's demonstration that very powerful deductions about linguistic systems can be made directly from psychological findings about human cognition makes their paper a most important and significant contribution to anthropology and linguistics. In order to clarify and strengthen their demonstration, the present paper offers a number of improvements of their deductive apparatus, some more general and incisive restatements of their assumptions, and two additional assumptions that are consistent with their original ones.

notes

[1] I would like to thank Hugh Gladwin and Martin Orans, for their very helpful discussions with me of the material presented here, and Lynn Thomas, for drawing the figures. I also want to express my deep gratitude to Judy Z. Kronenfeld for her careful readings of and her extensive editorial work on this paper.

[2] Bruner, Goodnow, and Austin actually speak of "conjunctive categories" rather than "conjunctively defined categories." They oppose these to "relational categories" and "disjunctive categories." What they call a "conjunctive category" is, in my classification, conjunctively defined by absolute attributes; what they call a "relational category" is conjunctively defined by a set of attributes containing at least one relative attribute. "Disjunctive categories" are disjunctively defined. I have restated their findings in this fashion because their statements obscure the common elements of their conjunctive and relational categories and because it seems to me generally preferable to distinguish between the characteristics of attributes—i.e. in this case "relative" vs. "absolute"—and the way in which attributes are used to define a category—i.e. "conjunctively" or "disjunctively."

[3] b. . .b represents both m. . .m and f. . .f but not m. . .f or f. . .m.

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